Paper 10.25 - Toolkit for the T10 Master Receipt

CQE-paper-10.25
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-10.25.md

Paper 10.25 - Toolkit for the T10 Master Receipt

Purpose

Paper 10.25 describes the tools for reviewing the T10 master receipt. These tools expose receipt binding, transport classification, local witness replay, and lookup cache materialization. They do not close the open lifts.

Review Objects

The toolkit works with:

```
Paper 00 binding
observer center C00
00 -> 1 encoding event
paper receipt bindings P01..P09
transport obligation rows
lookup receipts
open-lift set
master verdict
```

Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-10/verify_t10_master_receipt.py
production/formal-papers/CQE-paper-10/t10_master_receipt.json
```

Primary package bindings:

```
lattice_forge.transport_obligations.verify_transport_obligations
lattice_forge.cmplx_lookup_cache.build_default_cache
lattice_forge.cmplx_lookup_cache.LookupReceipt
```

The promoted verifier is the replayable route for the paper. The kernel notes currently mark `production/papers/CQE-paper-10/02-CQE-tool/run.py` as missing.

Analog Toolkit

A physical reconstruction can be done as an index notebook:

```
bind Paper 00 and Papers 01-09 as tabs
write the observer center C00 on the front tab
write the 00 -> 1 encoding event
attach each paper receipt path
write four transport rows
color-code classifications
copy lookup receipt counts
mark open lifts without erasing them
```

Suggested classification marks:

```
green  = demonstrated
amber  = bounded_local or bounded_external
blue   = registered_landing_forms
black  = open
```

Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

```
How many transport rows are in T10?
How many are demonstrated?
How many open or non-demonstrated lifts remain visible?
Does T10 prove a cold-start closed-form fingerprint map?
```

Expected answers:

```
4
2
2
no
```

Boundary

Paper 10.25 is a toolkit supplement. Any claim that T10 closes open lifts, cold-start maps, or later domain physics must pass Paper 10.50's claim contract before it can enter a scientific paper.